REAL ESTATE MARKET TRENDS DASHBOARD

Project README

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1. Project Overview

The **Real Estate Market Trends Dashboard** is an interactive Power BI project developed to analyze property-market data and present important market metrics in a clear, decision-oriented dashboard.

The dashboard helps users explore **property prices, property types, city-wise property distribution, bedroom categories and brokerage-related information**. It combines KPI cards, charts and interactive slicers to make large amounts of property data easier to understand.

2. Objectives

- Monitor overall property-market size and key KPIs.
- Analyze property distribution across different property types.
- Compare property volumes across cities.
- Compare average property prices across property types and cities.
- Analyze the market using price and bedroom-based filtering.
- Provide interactive filtering through slicers.
- Present real estate data in a simple and visually understandable format.

3. Tools & Technologies

Power BI Desktop - data modeling, DAX measures, interactive visuals, slicers and dashboard design.

Power Query - data preparation and transformation.

DAX - calculation of analytical measures and KPIs.

Real Estate Dataset - source data used for property-market analysis.

4. Key Performance Indicators

KPI	Displayed Value
Total Properties	Approx. 34K
Total Cities	17
Average Property Price	Approx. 5.12K
Total Property Value	Approx. 173.87M
Count of PRICE	Approx. 33.93K

Note: The price values and total property value are reported in the units displayed by the supplied dashboard. The screenshot does not explicitly specify the currency.

5. Interactive Filters

- **PROPERTY_TYPE** - filters the dashboard by selected property category.
- **CITY** - filters the analysis by selected city/locality.
- **PRICE** - enables analysis of selected property-price ranges.
- **BEDROOM_NUM** - filters properties according to bedroom count.
- **BROKERAGE** - enables brokerage-related filtering.

6. Dashboard Visuals

- **Property Price by Property Type** - compares price-record volume across property categories.
- **Total Property by CITY** - compares and ranks cities according to property volume.
- **Average Property Price by PROPERTY_TYPE** - compares average prices across property categories.
- **Property Type Distribution** - donut chart showing the share of properties by property type.
- **Total Properties by PROPERTY_TYPE** - compares the number of properties in each category.
- **Average Property Price by CITY** - ranks cities/localities according to average property price.

7. Key Insights

- The dashboard represents approximately **34K properties across 17 cities**.
- **Apartments** are the dominant property category, representing approximately **83.72%** of the displayed property distribution.
- **Gurgaon and Hyderabad** appear among the highest-volume cities in the displayed property-count ranking.
- **Service Apartments** show the highest displayed average property price among the visible property types, at approximately **13.6K**.

- Several **Mumbai localities**, including Mumbai South West and Mumbai Andheri-Dadar, appear near the top of the displayed average-price ranking.
- Interactive slicers allow users to investigate specific market segments rather than relying only on overall KPIs.

8. Repository Contents

- `Real_Estate_Market_Trends.pbix` - Power BI dashboard file.
- `Dataset` - source/cleaned real estate dataset used for analysis.
- `Screenshots` - final dashboard image(s).
- `Taniya_CodeAlpha_Real_Estate_Market_Dashboard_Report.pdf` - detailed project report.
- `Real_Estate_Market_Trends_README.md` - project overview and usage summary.

9. How to Use the Dashboard

1. Open the `.pbix` file using **Microsoft Power BI Desktop**.
2. Use the slicers at the top of the dashboard to select a property type, city, price range, bedroom number or brokerage.
3. Review the KPI cards for the selected market segment.
4. Compare property volume and average prices using the charts.
5. Clear or change filters to perform additional market comparisons.

10. Conclusion

The **Real Estate Market Trends Dashboard** converts property data into an interactive analytical view. It provides a single place to monitor market KPIs and investigate differences in property type, property volume, average price and city-level market patterns using slicer-based filtering.

This project demonstrates practical skills in **Power BI, data visualization, dashboard development, KPI analysis and business intelligence**, and was developed as part of the **CodeAlpha Internship**.

11. Future Scope

- Add time-based property price and volume trends when date information is available.
- Add map-based geographic analysis.
- Add price-per-square-foot analysis if area data is available.
- Add bedroom-wise and property-type-wise price analysis.
- Add brokerage performance analysis.
- Add predictive analytics and machine-learning-based price forecasting.
- Connect the dashboard to a regularly refreshed data source.

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